

Airbnb NYC Data Pipeline: From Raw Data to Revenue Insights



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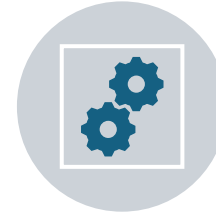
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Problem Definition

- **1. Problem Definition and Business Context**

- The Airbnb NYC dataset contains significant data quality challenges that prevent hosts from making informed business decisions. The raw data includes missing values, inconsistencies, corrupted text, invalid entries, and lacks direct revenue information. Most critically, landlords struggle to understand profitability drivers without clean, structured data. This project transforms messy Airbnb data into actionable insights that help hosts optimize pricing, policies, and operational strategies to maximize revenue in New York City's competitive short-term rental market.

- **Data Quality Challenges**

Missing values, inconsistent formatting (currency symbols, percentages), corrupted text in descriptions, invalid negative values in constrained fields

- **Missing Revenue Data**

No direct revenue column; estimated using price × reviews per month proxy metric

- **Decision Paralysis**

Hosts make decisions based on intuition rather than data-driven insights about demand drivers

- **Market Complexity**

Diverse neighborhoods, room types, and operational policies require segmented analysis

- **Competitive Pressure**

New York City's overcrowded market demands optimization to remain profitable

Data Engineering Workflow and Pipeline Architecture

- The project implements a reproducible, multi-stage data pipeline that systematically transforms raw data into analysis-ready datasets. The architecture follows a three-tier design separating raw, cleaned, and analytical data layers, ensuring data quality standards and enabling scalable maintenance.

PIPELINE STAGE	INPUT	OUTPUT	KEY TECHNIQUES
Ingestion	CSV file (102,599 rows)	Raw data table	Python, pandas → pd.read_csv(...)
Cleaning	Raw table with inconsistencies	Validated records	Duplicate removal, invalid value filtering, format standardization
Transformation	Cleaned table	Structured records	String-to-numeric conversion, binary encoding, conditional logic
Feature Engineering	Structured records	Enhanced features	Listing age, estimated revenue, stay categories, interaction features
Analysis	Feature-rich dataset	Aggregated insights	Correlation analysis, group-by aggregations, market segmentation
Reporting	Analytical outputs	Visualizations & tables	Distribution charts, revenue analysis, neighborhood ranking

Data Quality Issues and Resolution Strategies

Irrelevant Features Removed

ID columns (host_id, id), free-text fields (NAME, host_name), and invariant columns (country—all US listings) dropped as they provided no analytical variation or were unstandardized

Formatting Inconsistencies

Price and service fee columns contained "\$", "%", and comma symbols; standardized by removing special characters and converting to numeric format using regex operations

Invalid Values

Negative values in minimum_nights and availability_365 removed (7 rows); approximately 2,700 rows with availability_365 > 365 capped at 365 to preserve data while enforcing logical bounds

Missing Value Handling

License column (99.99% missing) dropped entirely; reviews_per_month imputed conditionally to avoid contradictions with number_of_reviews; numerical features filled with median; categorical features with mode

Duplicate Records

All duplicate rows removed to prevent inflated statistics

Coordinate Validation

Missing latitude/longitude rows dropped (8 rows) to prevent invalid geographic points

Feature Engineering and Data Transformation

- Feature engineering created new variables to fill analytical gaps and capture hidden patterns in the data. Derived features proxy for missing measurements (estimated revenue for actual bookings), temporal dynamics (listing age from construction year), and operational characteristics (host type classification). Transformation techniques included binary encoding, categorical binning, and interaction features to enhance predictive power.

- **Estimated Monthly Revenue**
 - Created as $\text{Price} \times \text{Reviews Per Month}$ to proxy actual booking revenue when direct data unavailable
- **Listing Age**
 - Derived from construction year (2026 - Construction Year) to capture temporal trends in listing maturity and market dynamics
- **Stay Categories**
 - Binned minimum_nights into Short-stay (1-3), Weekly (4-7), Monthly (8-30), Long-term (30+) for non-linear trend analysis
- **Host Type Classification**
 - Conditional rule based on calculated_host_listings_count to segment Individual vs. Professional hosts
- **House Rules Binary**
 - Transformed unstructured text into binary indicator (has_house_rules: 0/1) for analytical tractability
- **Days Since Last Review**
 - Converted date to days elapsed (reference: 2022-09-11) as recency proxy for listing activity
- **Log Transformations**
 - Applied to right-skewed features (number_of_reviews, reviews_per_month, calculated_host_listings_count) while preserving originals for interpretability

Key Revenue Drivers and Market Insights

- Analysis reveals that booking frequency dramatically outpaces all other factors in driving revenue. Reviews per month exhibits 0.83 correlation with estimated revenue—nearly double the next predictor. Geographical location, room type, and minimum stay requirements significantly influence demand patterns. Counter-intuitively, cancellation policies and instant booking features show minimal impact, while house rules negatively correlate with revenue.

Revenue Driver	Correlation/Finding	Business Implication	Priority
Reviews Per Month	0.83 correlation	Booking frequency is the dominant revenue lever; drives 3x revenue difference between tiers	Critical
Number of Reviews	0.52 correlation	Historical booking success provides credibility signal to future guests	High
Price	0.31 correlation	Higher nightly rates provide modest revenue lift but less than occupancy frequency	Medium
Minimum Nights	-0.08 correlation	Longer stay requirements actively suppress demand and revenue	Negative
Review Rate (Stars)	0.03 correlation	Star ratings have negligible impact on revenue—guest volume matters more	Low
Room Type	Categorical leader	Hotel rooms generate highest demand; private rooms second; entire homes serve niches	High
Neighborhood Group	Staten Island: \$1,000+ avg; Manhattan: \$300-400	Geographic clustering of high-value areas; micro-targeting more effective than broad borough spending	High

Actionable Business Recommendations for Hosts

- The analysis synthesizes data insights into concrete, prioritized actions that maximize revenue potential. Recommendations emphasize booking frequency optimization, strategic minimum stay policies, geographic arbitrage, and targeted marketing by room type. The insights challenge common assumptions (e.g., strict policies don't suppress demand) while confirming others (e.g., location matters deeply).

\$2,914

Maximize Booking Frequency (Highest Impact)

Set minimum nights to 1-3 nights; respond to inquiries rapidly; price competitively; actively request guest reviews. Increasing from 0 to 3+ reviews/month crosses from \$0 to \$2,914 estimated monthly revenue—single highest leverage point

0.31

Room Type Marketing

Emphasize hotel rooms for stable high-volume revenue; market entire homes on privacy/local experience angle; niche shared rooms toward budget social travelers. Do not eliminate lower-performing types—serve different segments

53%

Optimize Minimum Stay Requirements

Avoid setting minimum stays above 7 nights; short-stay focus (1-3 nights) captures 53% of market demand. Long-term rentals (30+ nights) show dramatically lower booking rates despite traditional assumptions

Chelsea

Neighborhood Prioritization

Invest in Staten Island (Chelsea, Huguenot highest performers) and Queens (Springfield Gardens, Glen Oaks); avoid broad Manhattan spending; micro-target high-revenue pockets within boroughs instead

53%

Strategic House Rules Review

Properties without house rules generate 53% of revenue; evaluate whether current rules create unnecessary booking friction. Test reducing restrictions to improve conversion

30+

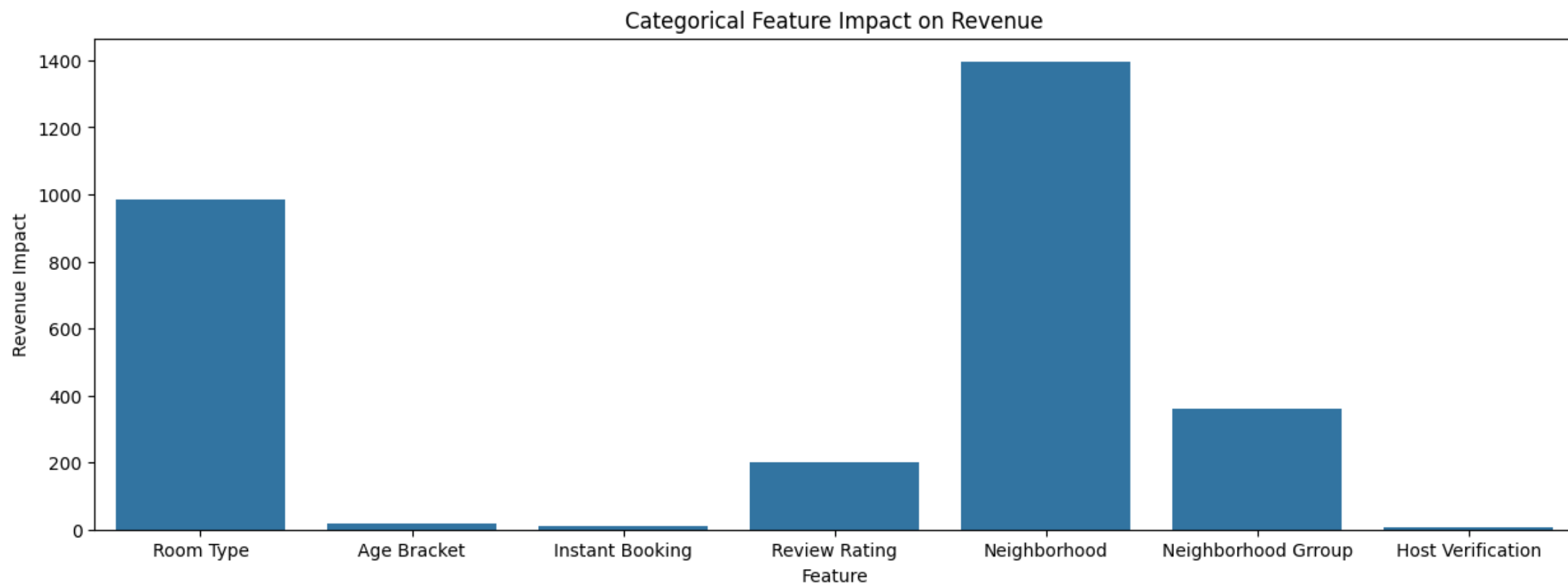
Price Strategically but Secondary

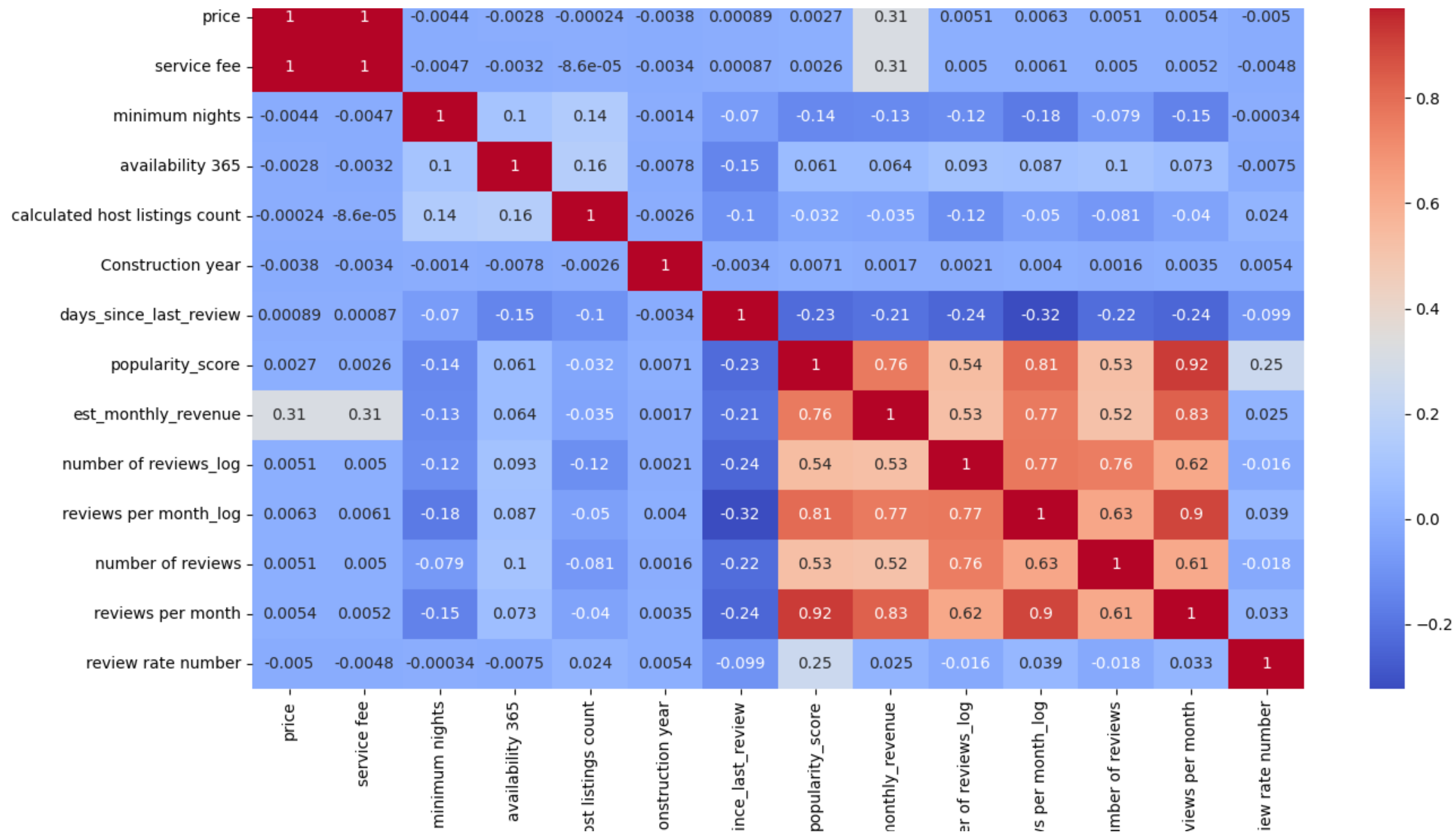
Raising prices alone generates minimal revenue lift (0.31 correlation). Use pricing to fill gaps and improve occupancy—occupancy frequency matters far more than nightly rate

Visualisation and reports

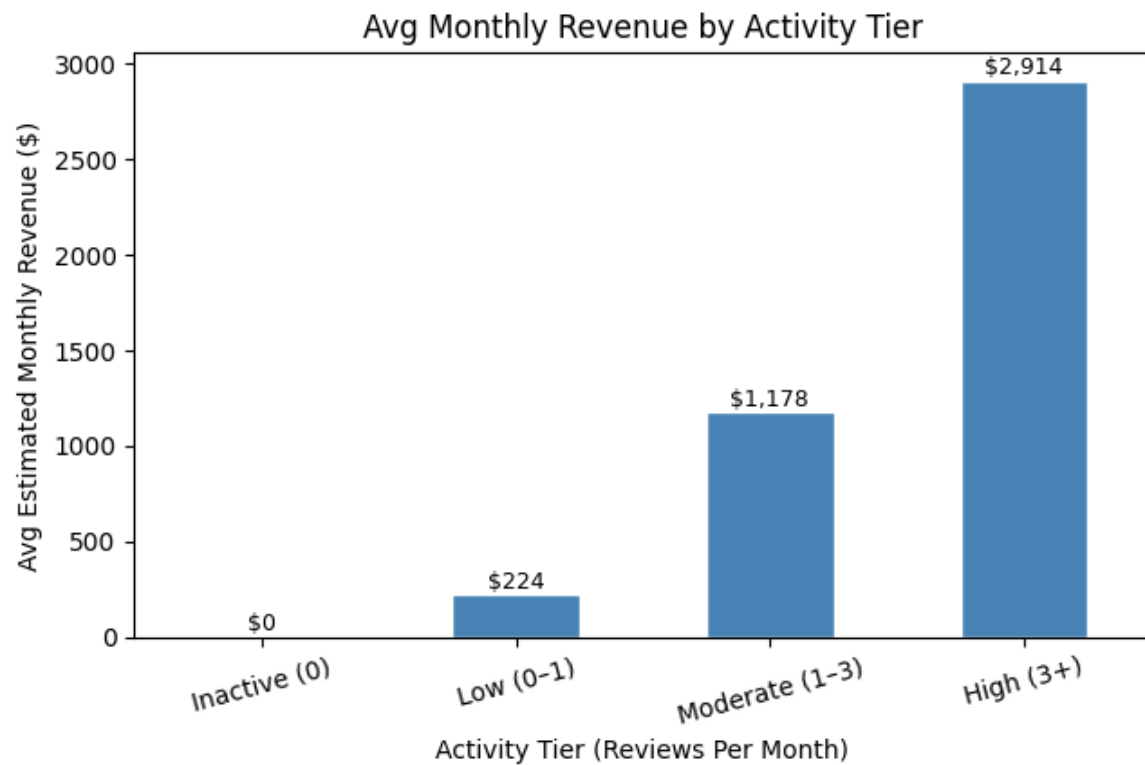
Numerical Feature Correlation with Revenue



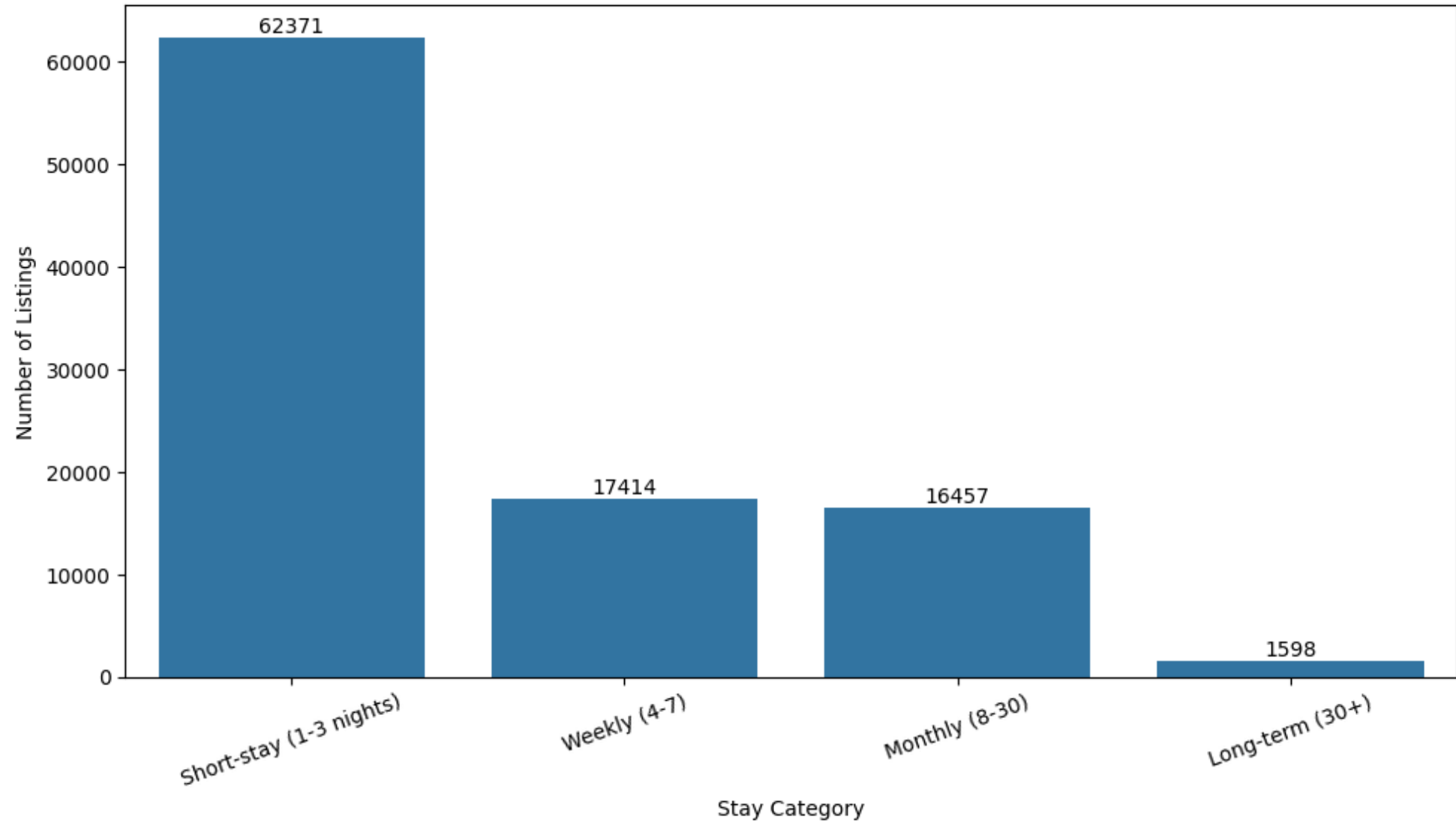


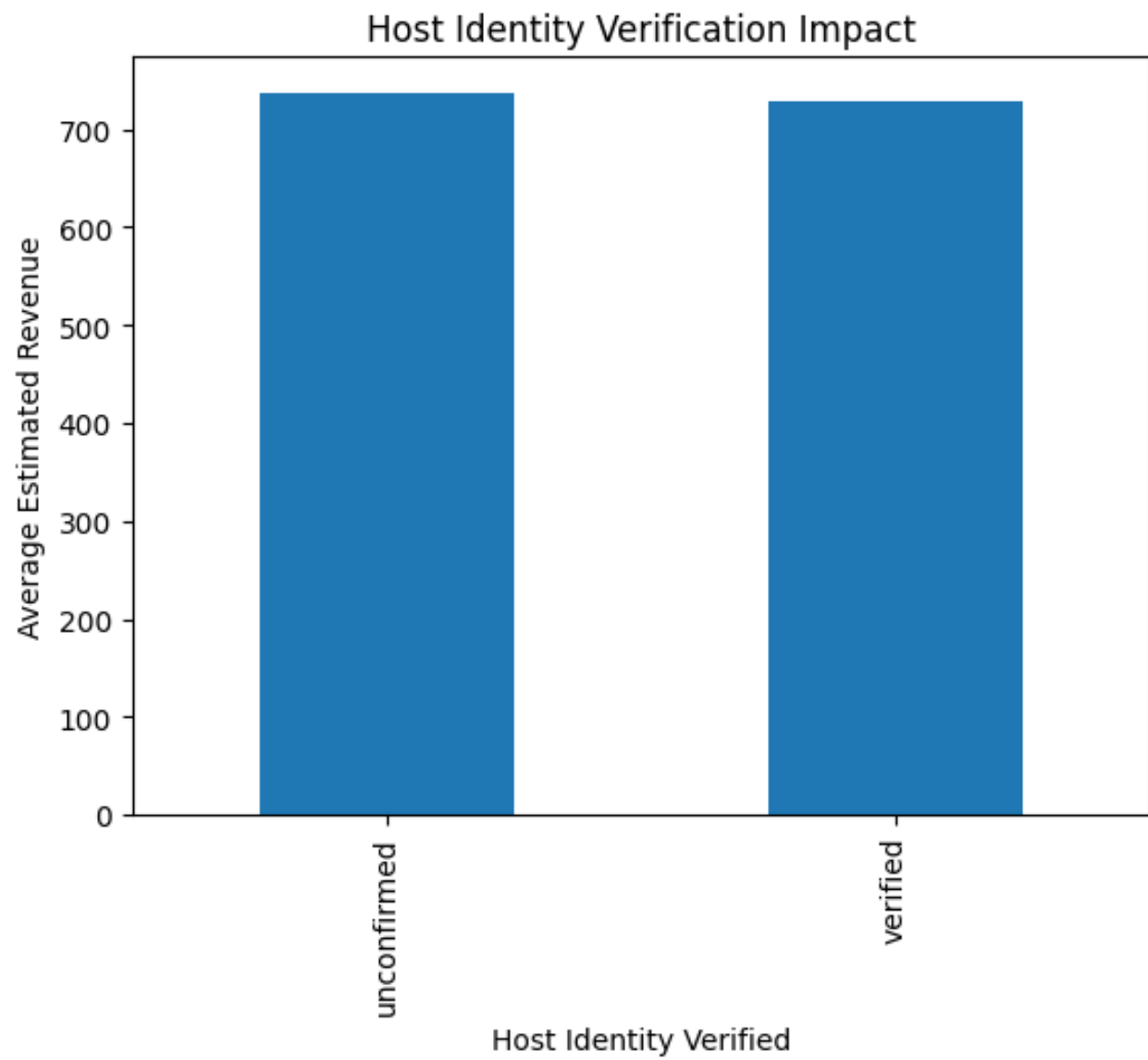


Reviews Per Month Impact on Estimated Revenue

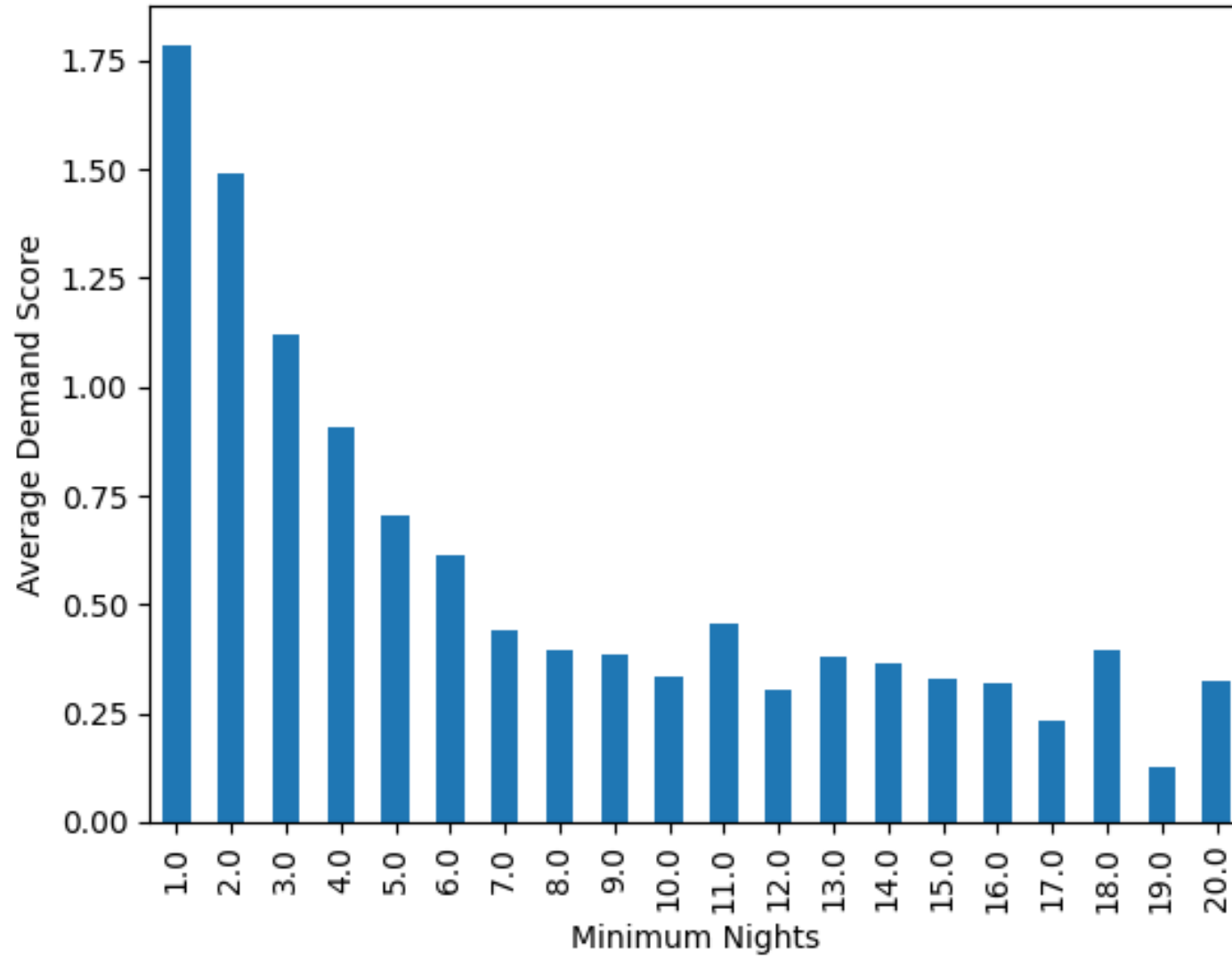


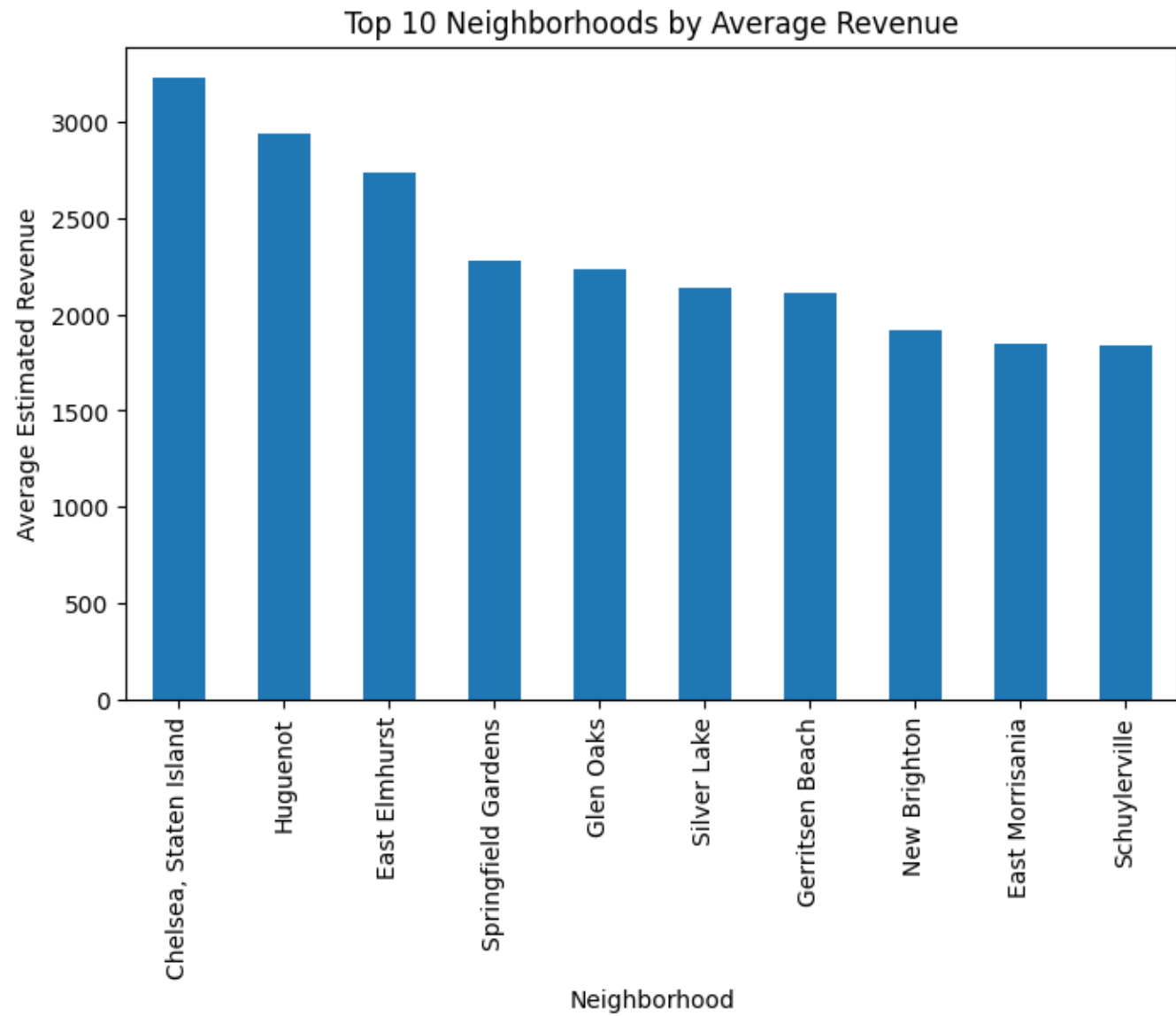
Distribution of Stay Categories

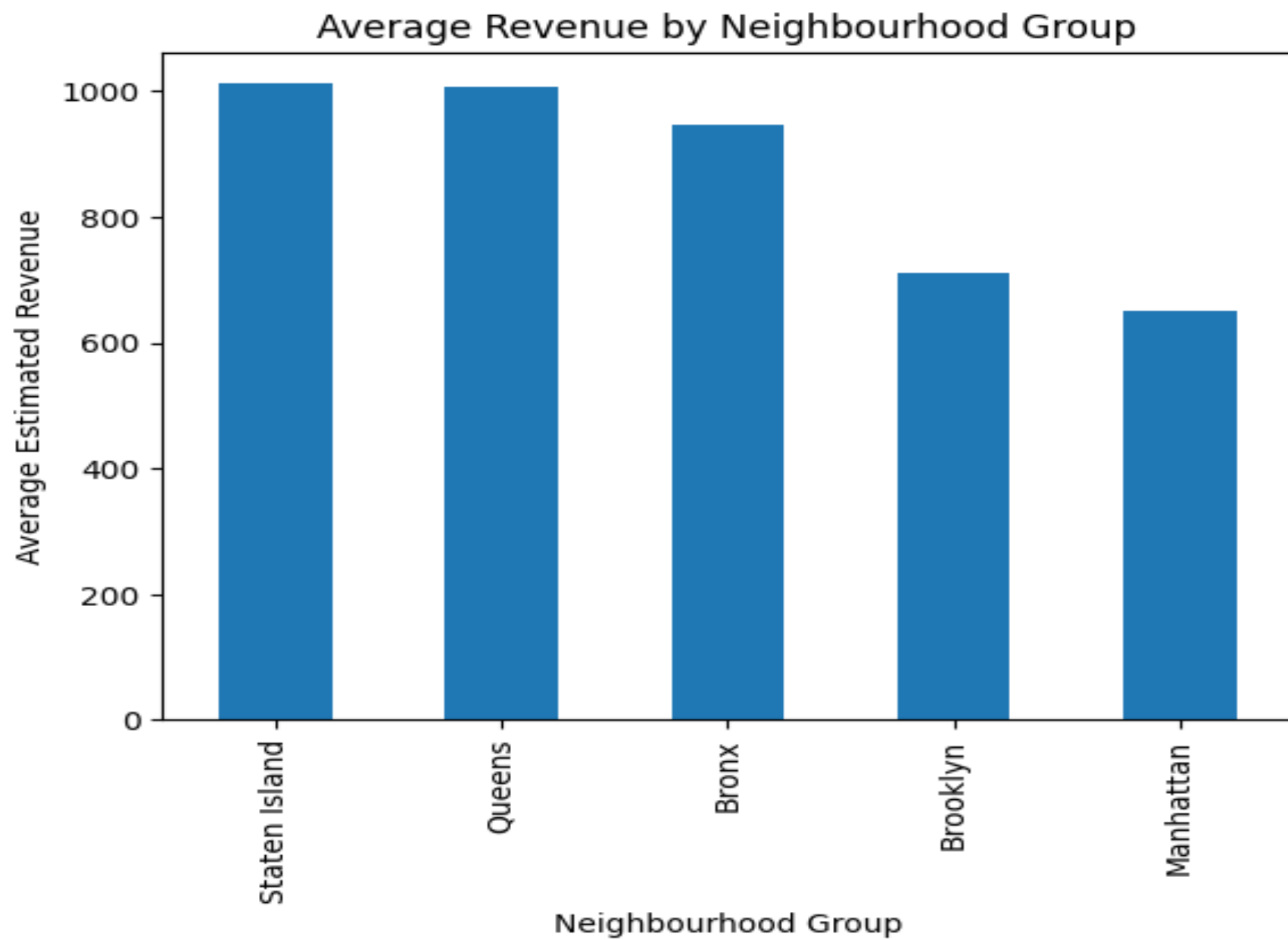


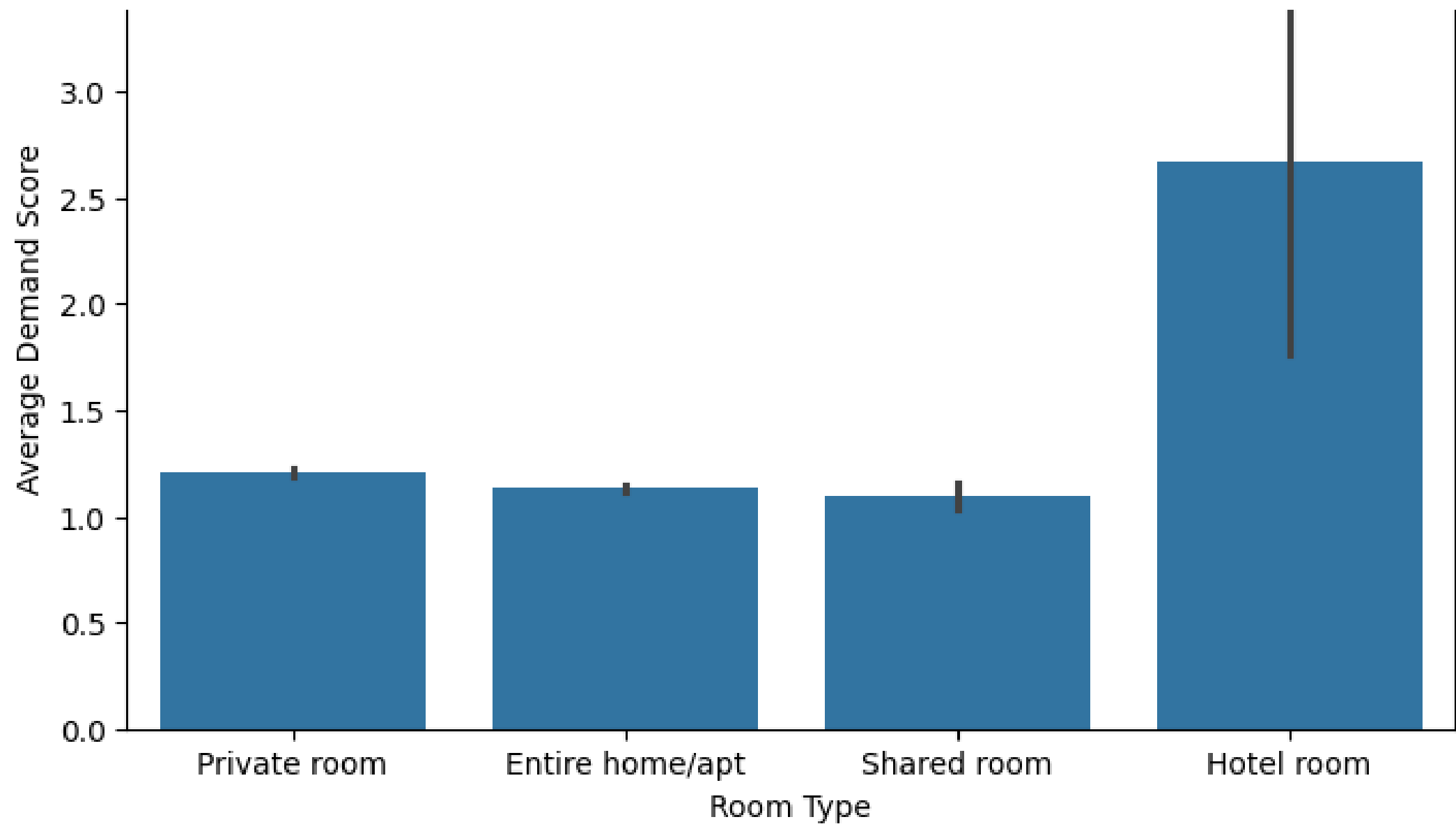


Demand by Minimum Nights









- Reposotry:

- https://github.com/nourswaid/Enginering_project.git

- Dataset:

- <https://www.kaggle.com/datasets/arianazmoudeh/airbnbopendata>
[a](#)